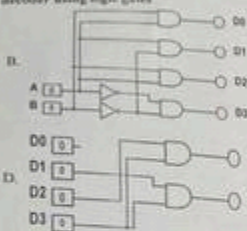


Active High decoder using logic gates



Output of D Flip flop



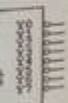
Q'



$(Y(E+D'))'$ will result in
+ED'

the SUM of a Full Adder.

- A. Connect Y1, Y2, Y4, Y7 with NAND
B. Connect Y1, Y2, Y4, Y7 with NOR
C. Connect Y0, Y3, Y5, Y6 with AND
D. Connect Y0, Y3, Y5, Y6 with NAND



	00	01	11	10
0	1	1	1	1
1				
2				
3	1			1

- A. $F = \bar{A}B + C\bar{D}$
B. $F = \bar{A}B + C\bar{D} + A\bar{B}$
C. $F = \bar{A}B + C\bar{D} + B\bar{D}$
D. $F = \bar{A}B + C\bar{D} + B\bar{C}$

16. The equation _____ cannot be further simplified.

- A. $\bar{A}B + A\bar{C} + \bar{B}C = x$
B. $\bar{A}B + AB + \bar{A}C = x$
C. $\bar{A}B(\bar{C} + C) = x$
D. $\bar{A}B + C = x$

17. The W output of the following circuit is



- A. $A + B$
B. $A > B$
C. $A < B$
D. $A - B$

18. The following truth table represents a _____ gate

X	Y	F
0	0	0
0	1	1
1	0	1
1	1	0

- A. XNOR
B. XOR
C. NAND
D. NOR

19. The output of SR flip flop when S=1, R=0 is _____

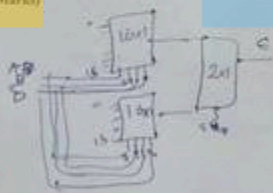
- A. 1
B. 0
C. No change
D. Not allowed

20. Which of the following will correctly complete the JK flip-flop truth table shown below?

J	K	Qn	Qn+1
0	0	0	?
0	1	?	?
1	0	?	?
1	1	?	?

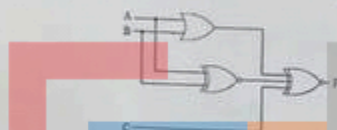
A	B	Q	C	D
1	1	0	1	1
1	1	1	1	0
0	0	0	0	1

Q2: Design a 32x1 multiplexer using two 16x1 multiplexers and a 2x1 multiplexer using block diagrams. (6 Marks)



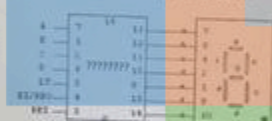
6. Choose the correct answer

- What is the output of XOR with two inputs A,B, if A is connected to +5v:
A. Always zero B. Always one C. B D. B'
- _____ is a combinational circuit that converts binary information from a input lines to a maximum of 2^n unique output lines
A. Adder B. Multiplexer C. Decoder D. Encoder
- To implement $A \oplus B$ using XOR with three inputs, the third input should be connected to:
A. Ground B. To one of inputs (A,B) C. +5V D. Answers b or c
- The sum of minterms of the simplified function $F(A,B,C,D) = B'D'$ is:
A. $\Sigma(4,6,12,14)$ B. $\Sigma(0,1,9,11)$ C. $\Sigma(0,2,8,10)$ D. $\Sigma(2,3,14,15)$
- For the output F to be 1 is the logic circuit shown, the input combination should be



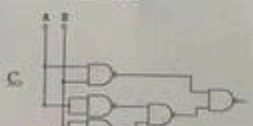
- A. $A=1, B=1, C=1$
B. $A=1, B=0, C=0$
C. $A=0, B=0, C=1$
D. $A=0, B=1, C=0$

6. The required decoder to drive the following 7-segment is



- A. 4884
B. 7448
C. 7447
D. 4774

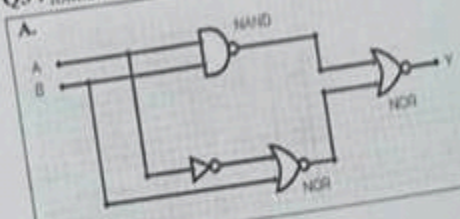
- The carry output of a half-adder circuit can be expressed as
A. $Cout = AB$ B. $Cout = A \oplus B$ C. $Cout = A + B$ D. None of these
- A multiplexor with a 4-bit data select input is a
A. 4:1 multiplexor B. 2:1 multiplexor C. 16:1 multiplexor D. 8:1 multiplexor
- Which one of the following figures shows the function $F = A \oplus B$ using NAND gates



10. The Boolean function $Y = AB + CD$ is to be realized using only 2 input NAND gates. The minimum number of gates required is
A. 2 B. 3 C. 4 D. 5

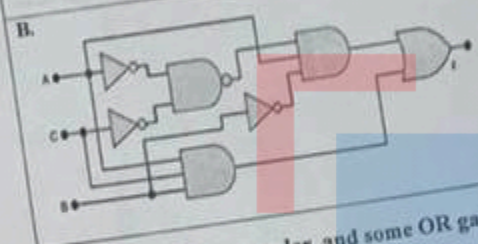
Course Title: Digital Logic and Design Lab
 Semester: 2nd
 Exam Date: 1/04/2024
 Student Name: _____
 GENERAL INSTRUCTIONS:
 1. Students are
 2. Late fee
 3. ...

Q3: Simplify a logic circuit: (5 Marks)



Final Answer:

$Y = AB$



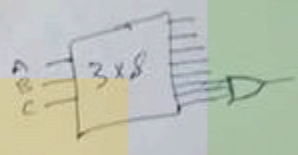
Final Answer:

$Z = ABC$

Q4: Given a 3-to-8 decoder, and some OR gates, draw the circuit for the following function: (5 Marks)

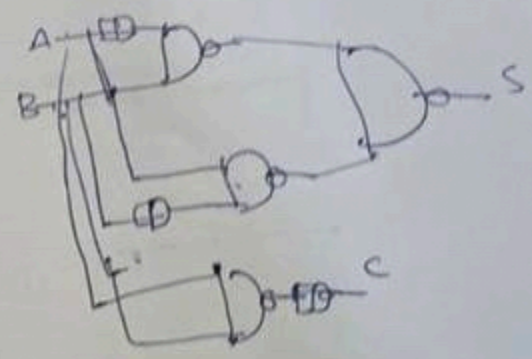
$F = AB + AB'C$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

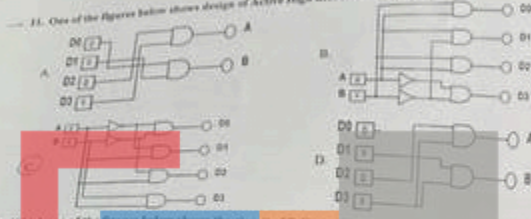


(4 Marks)

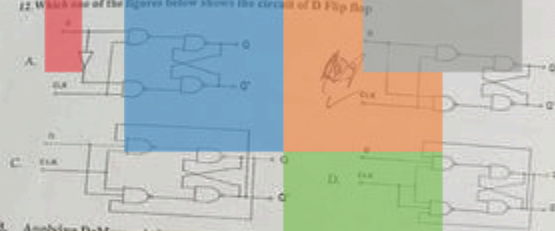
Q5: Draw the HALF ADDER circuit using:
 a) only NAND gates.
 b) OR BY USING only NOR gates



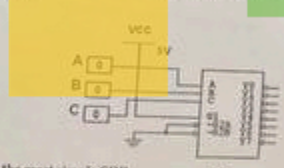
11. One of the figures below shows design of Active High decoder using logic gates



12. Which one of the figures below shows the circuit of D Flip flop



13. Applying DeMorgan's Law to $F = ((AB)' + C)'(E + D)'$ will result in
 A. $F = A'B'C + E + D$ B. $F = ABC + EDF$ C. $F = A'BC + (E + D)'$ D. $F = A + BC + D$
14. To complete the figure below to implement the SUM of a Full Adder, you should



- A. Connect Y1, Y2, Y3, Y7 to 0
 B. Connect Y1, Y2, Y3, Y7 to 1
 C. Connect Y0, Y3, Y5, Y7 to 0
 D. Connect Y0, Y3, Y5, Y7 to 1

15. Identify the most simple SOP expression from the K-map shown

	CD		00	01	11	10
AB	00	1	1	1	1	1
	01					
	11					
	10	1				

- A. $F = AB + CD$
 B. $F = AB + CD + ABC$
 C. $F = AB + CD + BD$
 D. $F = AB + CD + AC$

16. The equation $AB + AC + BC = X$

17. The W output of the following



18. The following truth table represents

X	Y	F
0	0	0
0	1	1
1	0	1
1	1	0

19. The output of SR flip flop when S=1, R=0

20. Which of the following will correct flip-flop truth table shown below?

J	K	Qn	Qn+1
0	0	0	?
0	1	1	?
1	0	1	?
1	1	1	?

Q2: Design a 32x1 multiplexer using 16x1 multiplexers (6 Marks)

